DETERMINATION OF INCOME AND EMPLOYMENT

Class XII Economics • Combined, detailed, exam-focused notes

Study approach

Concepts are explained first, followed by formulas, diagrams, policy measures, distinctions and exam-ready answers. The notes combine the structure of the existing detailed notes with additional topics found in the four uploaded class-note PDFs.

What these notes cover	Quick revision focus
Basic terms and income determination	AD, AS, effective demand, ex-ante/ex-post
Consumption, saving and investment	Functions, propensities, multiplier and numericals
Equilibrium and employment	AD–AS and S–I approaches; full and underemployment equilibrium
Employment and unemployment	Labour force, employment, unemployment and types
Demand gaps and corrective policies	Excess/deficient demand, fiscal and monetary measures
Paradox of thrift and exam practice	Concept explanation, formula sheet, diagrams and sample answers

Source-use note: The four class-note PDFs are scanned. Their readable content was interpreted with OCR; unclear handwriting or diagram labels have not been treated as certain. This combined version preserves the topics and terminology that could be identified and uses the existing detailed notes as the style and structure guide.

1. Meaning and Basic Terms

Aggregate Demand (AD)

Aggregate Demand is the total planned expenditure on final goods and services in an economy during a given period. In the simple two-sector model, households consume and firms invest.

$$AD = C + I$$

- C = consumption expenditure; I = investment expenditure.
- In an economy with government and foreign trade, AD may also include government expenditure and net exports.

Aggregate Supply (AS)

In the simple income-determination model, Aggregate Supply is represented by the value of planned output, equal to national income. Income is either consumed or saved.

$$AS = C + S = Y$$

Ex-ante and Ex-post

Term	Meaning	Use in the model
Ex-ante	Planned or intended consumption, saving or investment, before the activity takes place.	Compare planned AD and AS to determine equilibrium.
Ex-post	Actual or realised values measured after the activity has taken place.	Actual saving and investment are equal as an accounting outcome.

Effective Demand

Effective demand is the level of aggregate demand that determines equilibrium output and income. In the Keynesian model, output adjusts until planned aggregate demand equals aggregate supply.

Equilibrium condition: AD = AS

Investment

Investment expenditure means spending on capital goods that add to productive capacity, such as machinery, equipment and buildings. In the simple Keynesian model, investment is often treated as autonomous, meaning independent of current income.

2. Consumption Function

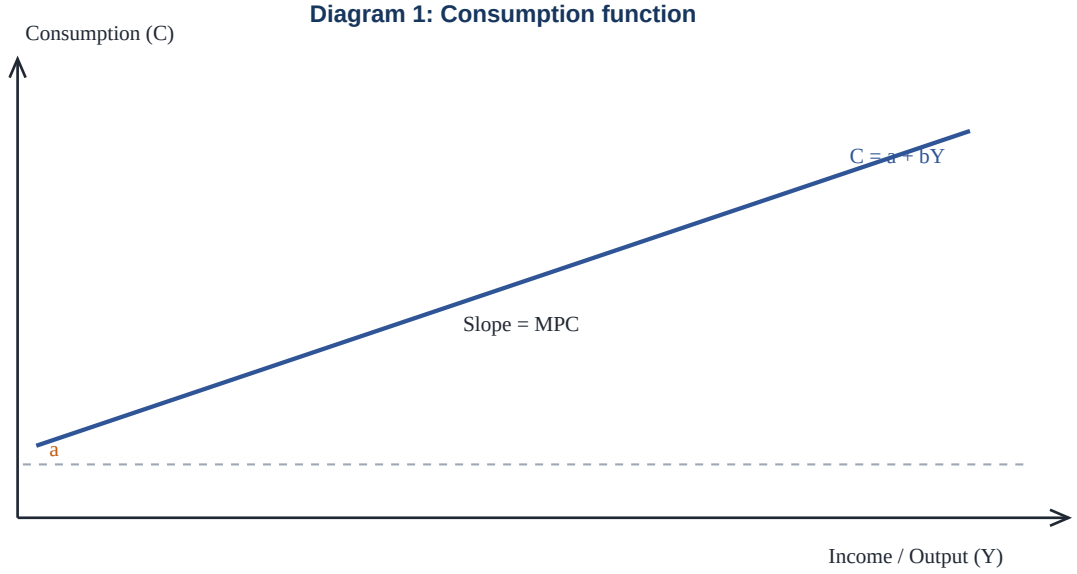
The consumption function shows the relationship between consumption expenditure and income, assuming other factors remain unchanged. Consumption usually rises as income rises.

$$C = a + bY$$

- **a — Autonomous consumption:** consumption that occurs even when income is zero; it may be financed from past savings or borrowing.
- **b — MPC:** the proportion of additional income spent on additional consumption.
- **Y — Income:** income in the model.

The consumption curve slopes upward. Its vertical intercept is autonomous consumption (a), and its slope is MPC (b).

Diagram 1: Consumption function



Average Propensity to Consume (APC)

$$APC = C / Y$$

APC measures the proportion of total income spent on consumption. It is an average ratio, not the change caused by a change in income.

Marginal Propensity to Consume (MPC)

$$MPC = \Delta C / \Delta Y$$

MPC measures the change in consumption divided by the change in income. In the simple linear consumption function, MPC is the slope of the consumption curve.

- In the basic model, MPC is generally between 0 and 1.
- APC may fall as income rises if consumption increases less than proportionately to income.
- Do not confuse APC (total C ÷ total Y) with MPC (change in C ÷ change in Y).

3. Saving Function

Saving is the part of income that is not spent on consumption. In the simple model, income is either consumed or saved.

$$S = Y - C$$

Substituting C = a + bY gives the saving function:

$$S = -a + (1 - b)Y$$

Average Propensity to Save (APS)

$$APS = S / Y$$

APS shows the proportion of total income saved.

Marginal Propensity to Save (MPS)

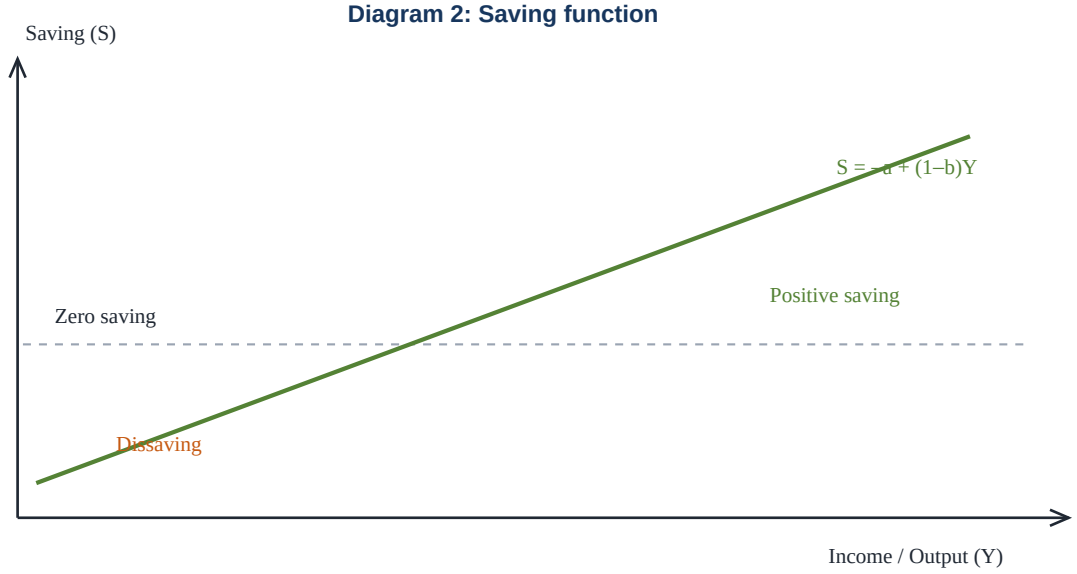
$$MPS = \Delta S / \Delta Y$$

MPS measures the part of additional income that is saved.

$$APC + APS = 1 \mid MPC + MPS = 1$$

These relationships follow because income is divided between consumption and saving. An additional unit of income is either consumed or saved in the simple model.

Diagram 2: Saving function



- When consumption is greater than income, saving is negative; this is called **dissaving**.
- At break-even income, consumption equals income and saving is zero.
- Above break-even income, income exceeds consumption and saving is positive.

4. Investment Multiplier

The investment multiplier explains why a change in autonomous investment can cause a larger change in equilibrium income. One person's expenditure becomes another person's income. The recipient spends part of the extra income, creating further rounds of expenditure.

$$K = \Delta Y / \Delta I$$

$$K = 1 / MPS = 1 / (1 - MPC)$$

- K = multiplier; ΔY = total change in income; ΔI = initial change in investment.
- Higher MPC → lower MPS → larger multiplier.
- Higher MPS → smaller multiplier because more of each additional income round is saved.

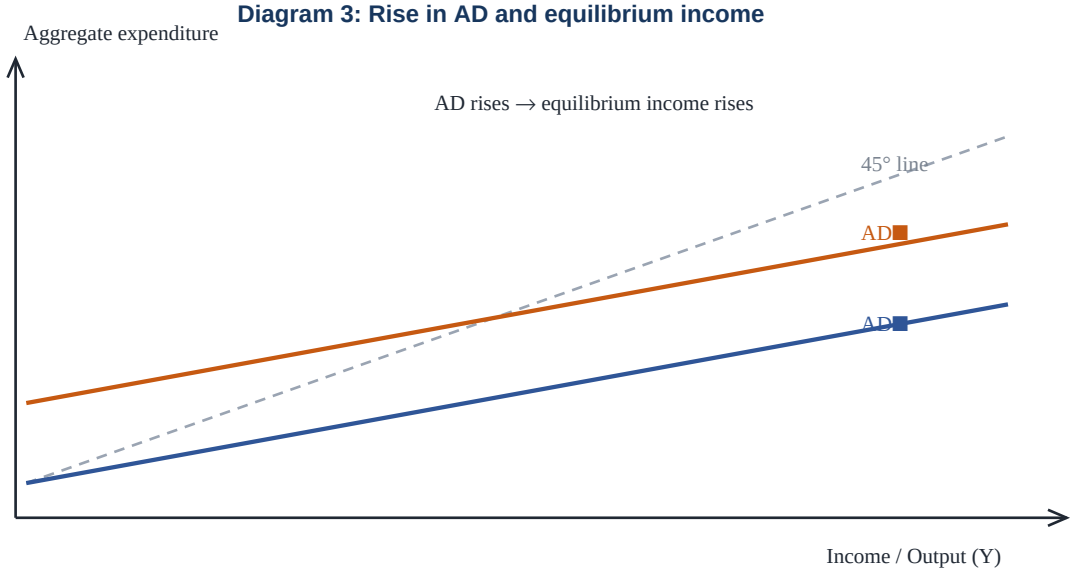
How to solve multiplier numericals

- Write the formula first.
- Identify whether the question gives MPC, MPS, ΔI or ΔY.
- Use K = 1/MPS or K = 1/(1–MPC) to find the multiplier.
- Use ΔY = K × ΔI when the multiplier and investment change are given.
- Show substitution and the final answer with the correct unit.

Multiplier process in words

An initial rise in investment increases income. Recipients spend a part of the additional income, which becomes income for others. This process continues in smaller rounds until the total income change is a multiple of the initial investment change.

Diagram 3: Rise in AD and equilibrium income



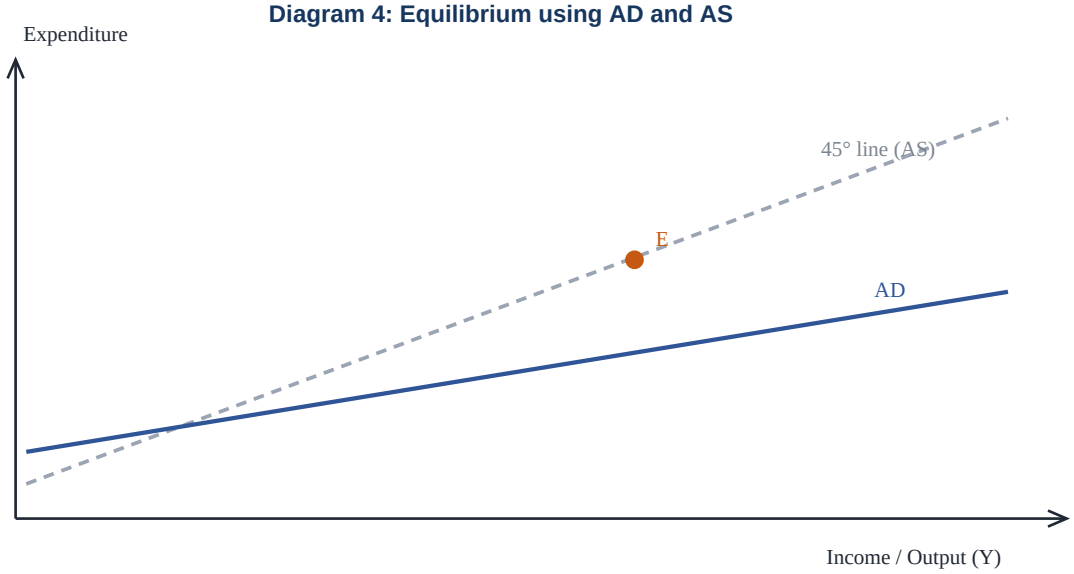
5. Equilibrium Level of Income

A. AD–AS approach

Equilibrium income is determined where planned aggregate demand equals aggregate supply. In the diagram, the AD curve intersects the 45-degree line, which represents points where output/income equals expenditure.

Equilibrium condition: AD = AS

Diagram 4: Equilibrium using AD and AS



- At intersection E, planned expenditure equals output.
- If planned AD is greater than output, inventories fall unexpectedly; firms are encouraged to raise production.
- If planned AD is less than output, unsold stocks accumulate; firms tend to reduce production.
- At equilibrium, there is no unintended change in inventories.

B. Saving–Investment (S–I) approach

In the two-sector model, equilibrium occurs where planned saving equals planned investment. This gives the same equilibrium income as the AD–AS approach.

Equilibrium condition: S = I

- If planned saving exceeds planned investment, planned expenditure is too low relative to output and income tends to fall.
- If planned investment exceeds planned saving, planned expenditure is relatively high and income tends to rise.
- At equilibrium, planned leakages through saving match planned investment injections.

Underemployment equilibrium

An economy can reach equilibrium even when some labour and other resources remain unemployed. If planned AD is insufficient to purchase the output associated with full employment, equilibrium income may be below full-employment income.

Key exam point: Equilibrium does not automatically imply full employment.

6. Employment, Unemployment and Full Employment

Labour force

The labour force is the part of the population that is willing and capable of working. The class notes distinguish the working/employed population from the unemployed population within the labour force.

Employment and unemployment

- **Employment:** a situation in which people are willing and capable of working and get a job.
- **Unemployment:** a situation in which people are willing and capable of working but do not get a job.

Types of unemployment

Type	Meaning
Voluntary unemployment	A person does not get or take the work of their own choice; the class notes describe this as not accepting work of one's choice.
Involuntary unemployment	A person is willing and capable of working even at the prevailing minimum wage rate but does not get a job.

Full employment

Full employment refers to a situation in which the whole labour force is employed and the resources of the economy are fully utilised. In the class-note framing, it also means zero involuntary unemployment.

Full-employment equilibrium

Full-employment equilibrium is the situation in which AD equals AS and the economy's resources are fully utilised at the full-employment level of output.

Underemployment equilibrium

Underemployment equilibrium occurs when AD equals AS, but the economy's resources are not fully utilised and some people remain involuntarily unemployed.

Full-employment equilibrium	Underemployment equilibrium
AD = AS at the full-employment output level.	AD = AS at an output level below full employment.
Resources are fully utilised.	Resources are not fully utilised.
No involuntary unemployment in the simplified definition.	Involuntary unemployment can exist.

Changes in equilibrium income

- An increase in autonomous consumption or investment shifts AD upward and raises equilibrium income.
- A decrease in autonomous consumption or investment shifts AD downward and lowers equilibrium income.

- Because of the multiplier process, the total change in income may be greater than the initial change in autonomous expenditure.

Adjustment through inventories

When output exceeds planned expenditure, unintended inventories rise and firms reduce output. When planned expenditure exceeds output, inventories fall and firms increase output. This adjustment continues toward equilibrium.

7. Excess Demand and Deficient Demand

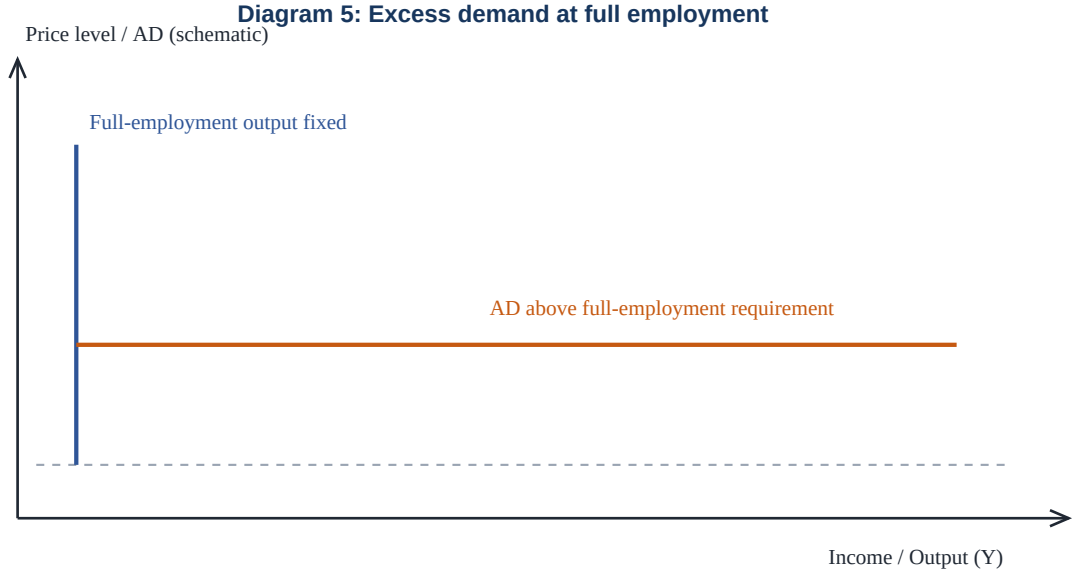
Excess demand / Inflationary gap

Excess demand exists when aggregate demand is greater than the level of demand required for full-employment equilibrium. The class notes describe this as AD exceeding the level consistent with full employment.

- Main impact: the general price level rises, creating inflationary pressure.
- At full employment, output, employment and income are treated as remaining constant in the standard diagram because the economy is already at its full-employment output.
- Possible causes include a rise in consumption, investment or government expenditure; a reduction in taxes; or an expansion of credit/money supply.

Corrective direction: reduce aggregate demand (AD).

Diagram 5: Excess demand at full employment



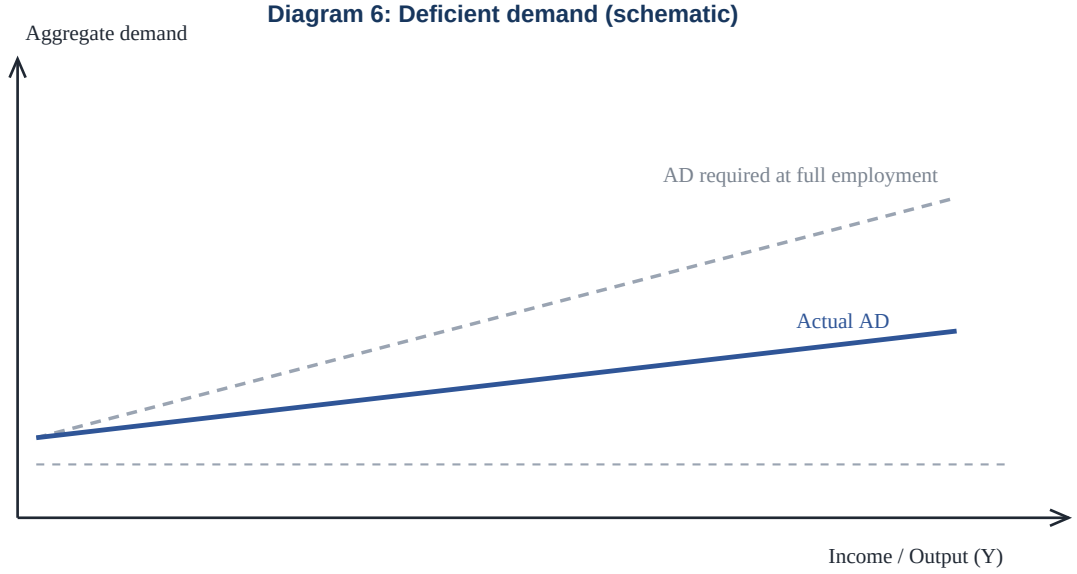
Deficient demand / Deflationary gap

Deficient demand exists when aggregate demand is below the level required to achieve full-employment equilibrium. In the class notes, the deflationary gap is the difference between the aggregate demand required at full employment and the lower actual planned aggregate demand.

- Impact: the price level may fall (deflationary pressure).
- Output, employment and income may fall because firms face insufficient demand.
- Possible causes include a fall in consumption or investment, higher taxes, reduced government spending, or contraction of credit.

Corrective direction: increase aggregate demand (AD).

Diagram 6: Deficient demand (schematic)



Excess demand vs deficient demand

Feature	Excess demand / inflationary gap	Deficient demand / deflationary gap
Position of AD	AD is above the level required for full employment.	AD is below the level required for full employment.
Price pressure	Upward pressure; inflation may result.	Downward pressure; deflation may result.
Output and employment	At full employment, output cannot rise much in the simplified model.	Output, employment and income may decline.
Policy direction	Reduce AD.	Increase AD.

8. Measures to Correct Excess and Deficient Demand

The class notes group corrective measures into fiscal measures and monetary measures. Monetary measures are further divided into quantitative and qualitative measures.

A. Fiscal policy measures

Measure	To correct excess demand	To correct deficient demand
Government expenditure	Reduce public/government expenditure to lower AD.	Increase public/government expenditure to raise AD.
Taxation	Increase taxes to reduce disposable income and spending.	Reduce taxes so people can spend more and AD rises.
Public borrowing / public debt	Restrictive fiscal action can help reduce spending demand, depending on the policy mix.	Expansionary spending and fiscal action can support demand.

Remember the direction: excess demand → reduce AD; deficient demand → increase AD.

B. Monetary policy measures

Monetary policy works through the money supply, availability of credit and borrowing costs. The class notes identify quantitative measures and qualitative measures.

Quantitative measures

Instrument	To correct excess demand	To correct deficient demand
CRR (Cash Reserve Ratio)	Increase CRR → banks have less lendable funds → credit and money supply fall → AD falls.	Decrease CRR → more lendable funds → credit and money supply rise → AD rises.
SLR (Statutory Liquidity Ratio)	Increase SLR to restrict the funds available for lending.	Decrease SLR to support greater lending.
LRR / reserve ratio	Increase the reserve requirement to restrict credit.	Decrease the reserve requirement to expand credit.
Bank rate	Increase the bank rate, making borrowing costlier and restraining credit.	Decrease the bank rate, making borrowing cheaper and encouraging credit.
Repo rate	Increase repo rate to make central-bank borrowing costlier and restrain credit.	Decrease repo rate to make central-bank borrowing cheaper and encourage credit.
Reverse repo rate	Raise the rate to encourage banks to park funds with the central bank, reducing funds available for lending.	Lower the rate to reduce the incentive to park funds and encourage lending.
Open market operations	Sell government securities; money is withdrawn from the economy and AD falls.	Buy government securities; money is injected into the economy and AD rises.

Qualitative measures

Instrument	To correct excess demand	To correct deficient demand
Margin requirement	Increase the margin requirement. Loans become harder to obtain against collateral, discouraging borrowing and reducing AD.	Decrease the margin requirement. Borrowing against collateral becomes easier, encouraging credit and increasing AD.
Moral suasion	The central bank persuades/pressures commercial banks to reduce the flow of credit.	The central bank persuades/pressures commercial banks to expand the flow of credit.
Selective credit control	Restrict credit for selected purposes or sectors to curb excessive demand.	Ease credit restrictions for selected purposes or sectors to support demand.

Fast memory aid

Excess demand: tighten policy → money/credit ↓ → AD ↓

Deficient demand: ease policy → money/credit ↑ → AD ↑

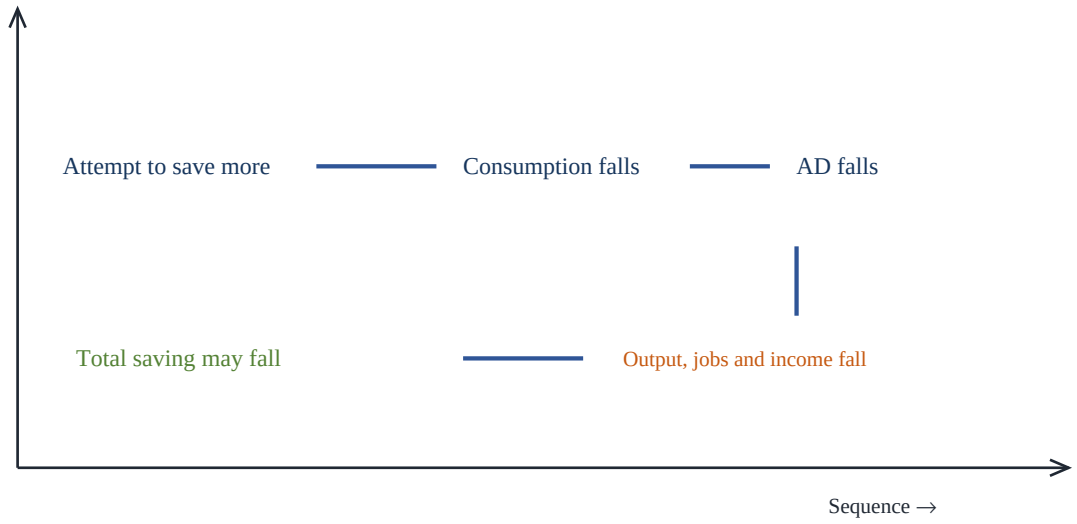
For exam answers, name the measure and explain the chain of effect on credit/money supply and aggregate demand.

9. Paradox of Thrift

The paradox of thrift is an idea associated with John Maynard Keynes. Saving may be beneficial for an individual, but if everyone tries to save more at the same time, total consumption can fall. Lower consumption reduces aggregate demand, which can lower production, employment and national income.

As national income falls, the total amount saved across the economy may also fall, even though people initially tried to save a larger share of their income.

Diagram 7: Paradox of thrift (cause-and-effect chain)



Why is it called a paradox of thrift?

Saving more may appear beneficial to each individual. However, if all households reduce consumption simultaneously, aggregate demand and national income can fall. Since total saving depends partly on total income, the economy may end up with lower total saving. This is why individually sensible behaviour can produce an undesirable result for the economy as a whole.

10. Formula Sheet

Concept	Formula / condition
Aggregate demand (two-sector model)	AD = C + I
Aggregate supply	AS = C + S = Y
Consumption function	C = a + bY
Saving function	S = Y – C = –a + (1 – b)Y
Average propensity to consume	APC = C / Y
Average propensity to save	APS = S / Y
Marginal propensity to consume	MPC = ΔC / ΔY
Marginal propensity to save	MPS = ΔS / ΔY
Average propensities	APC + APS = 1
Marginal propensities	MPC + MPS = 1
Investment multiplier	K = ΔY / ΔI = 1/MPS = 1/(1–MPC)
Change in income	ΔY = K × ΔI
Equilibrium conditions	AD = AS or S = I

Symbols

C = consumption; S = saving; I = investment; Y = income/output; a = autonomous consumption; b = MPC in the linear consumption function; K = multiplier; Δ = change.

11. Common Distinctions

Pair	Difference
APC vs MPC	APC = total consumption ÷ total income. MPC = change in consumption ÷ change in income.
APS vs MPS	APS = total saving ÷ total income. MPS = change in saving ÷ change in income.
Ex-ante vs Ex-post	Ex-ante values are planned/intended; ex-post values are actual/realised.
AD–AS vs S–I approach	AD–AS: planned AD = AS. S–I: planned saving = planned investment. Both give the same equilibrium income in the simple two-sector model.
Full vs underemployment equilibrium	Both may satisfy AD = AS. Full-employment equilibrium uses all resources; underemployment equilibrium occurs with resources left idle.
Excess vs deficient demand	Excess demand is above the full-employment requirement and causes inflationary pressure; deficient demand is below it and can reduce output/employment.

12. Exam-Ready Answers

1. Why is AD equal to AS at equilibrium?

At equilibrium, planned aggregate expenditure equals planned output. There is no unintended change in inventories, so firms have no reason to change production.

2. Why can equilibrium occur below full employment?

Aggregate demand may be insufficient to purchase the output associated with full employment. AD and AS can still intersect at a lower output level, creating underemployment equilibrium.

3. What is the investment multiplier?

The investment multiplier is the ratio of the total change in equilibrium income to the initial change in investment. It operates because additional spending creates income and further consumption in successive rounds.

4. Why does a higher MPC increase the multiplier?

A higher MPC means a larger part of each additional income round is spent. This creates more subsequent rounds of expenditure, so the total change in income is larger.

5. What is deficient demand?

Deficient demand occurs when aggregate demand is less than the level required for full-employment equilibrium. It may reduce output and employment and create deflationary pressure.

6. How can deficient demand be corrected?

Expansionary fiscal measures, such as higher government expenditure or lower taxes, and suitable expansionary monetary measures can raise aggregate demand.

7. How can excess demand be corrected?

Excess demand can be corrected by reducing aggregate demand. The government may reduce public expenditure or increase taxes. The central bank may raise reserve ratios or policy rates, sell government securities, or use selective credit controls to restrict credit.

8. Distinguish between full-employment and underemployment equilibrium.

In full-employment equilibrium, AD equals AS at the full-employment output level and resources are fully utilised. In underemployment equilibrium, AD equals AS at a lower output level and some labour or other resources remain unemployed.

9. What are the effects of deficient demand?

Deficient demand can lead to lower prices or deflationary pressure, reduced production, lower employment and a fall in income.

10. What is the paradox of thrift?

If households collectively try to save more by reducing consumption, aggregate demand may fall. Lower demand can reduce output, employment and income, so total saving in the economy may fall rather than rise.

13. Diagram Practice Checklist

- Consumption function: label Income (Y) and Consumption (C); show positive intercept a and upward slope b.
- Saving function: label Income (Y) and Saving (S); mark the zero-saving line, dissaving region and positive-saving region.
- AD–AS equilibrium: draw the 45-degree line, AD curve, intersection E and equilibrium income.
- S–I equilibrium: draw saving curve and investment line/curve; mark the intersection where S = I.
- Multiplier: show initial AD and higher AD after investment rises; indicate higher equilibrium income.
- Excess demand: show AD above the level required at full employment and explain inflationary pressure.
- Deficient demand: show AD below the level required for full employment and label the demand gap.
- Every diagram should have both axes labelled, all curves labelled, the equilibrium point and direction of change.

14. Quick Self-Test

Try answering these without looking at the notes. The answers are given below for quick checking.

- 1. Write the formula for MPC and MPS.
- 2. What is the equilibrium condition in the AD–AS approach?
- 3. What is the equilibrium condition in the S–I approach?
- 4. If MPC rises, what happens to the multiplier?
- 5. Which direction should AD move to correct deficient demand?
- 6. Name two fiscal measures to correct excess demand.
- 7. What happens to money supply when the central bank sells government securities?
- 8. Define underemployment equilibrium.

Answer key

1. MPC = $\Delta C/\Delta Y$; MPS = $\Delta S/\Delta Y$. 2. AD = AS. 3. S = I. 4. The multiplier rises. 5. AD should increase. 6. Reduce government expenditure and/or increase taxes. 7. Money supply decreases. 8. AD = AS at an output level below full employment, with some resources not fully utilised.

15. Final Revision Plan

- Learn the meanings of AD, AS, effective demand, investment multiplier and underemployment equilibrium.
- Memorise the formula sheet and the meaning of every symbol.
- Practise the consumption and saving functions, equilibrium diagrams, and demand-gap diagrams.
- Remember the policy direction: excess demand → reduce AD; deficient demand → increase AD.
- Prepare the short distinctions and the exam-ready answers.
- For numericals, practise one representative question for each formula type and show formula, substitution and final unit.

End of combined notes • Prepared in the same clean, numbered, exam-focused style as the supplied detailed notes.